

Jakub Klencki

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massive binary stars – stellar interactions and transients – gravitational-wave sources

EMPLOYMENT

10/2024 – now	MPA Research Fellow	<i>Max Planck Institute for Astrophysics</i>
10/2021 – 10/2024	ESO Research Fellow	<i>European Southern Observatory</i>

EDUCATION

09/2017 – 10/2021	PhD in Astrophysics (Cum Laude) <i>Massive binary stars and gravitational-wave sources (thesis)</i> advisors: Gijs Nelemans, Alina Istrate, Onno Pols	<i>Radboud University</i>
09/2015 – 09/2017	MSc. in Astronomy <i>Astrophysical origin of gravitational-wave sources</i> advisor: Chris Belczynski	<i>University of Warsaw</i>
09/2012 – 09/2015	BSc. in Physics <i>Machine learning: Rapid transient detection in photometric surveys</i> advisor: Łukasz Wyrzykowski	<i>University of Warsaw</i>

INVITED TALKS

2025	University College Cork	<i>EAS Symposium "Binary Stellar Interactions and Their Outcomes"</i>
–	University College Cork	<i>EAS Special Session on High-mass X-ray Binaries</i>
–	MPIA, Heidelberg	<i>Seminar</i>
2024	Niels Bohr Institute	<i>workshop "New Ideas on the Origin of Black Hole Mergers"</i>
–	Lorentz Center	<i>workshop "Benchmarking Stripped Stars and Their Companions"</i>
–	IAU G2 Comission	<i>Seminar</i>
–	MPA, Garching	<i>workshop "Challenges in Collisional N-body Dynamics"</i>
2023	Nicolaus Copernicus University	<i>XLI Meeting of Polish Astronomical Society</i>
–	Charles University in Prague	<i>Colloquium</i>
–	Geneva Observatory	<i>Seminar</i>
–	Lorentz Center	<i>discussion lead, "Renaissance of Stellar Black-Holes (...)"</i>
–	Warsaw Observatory	<i>Colloquium</i>
–	Potsdam University	<i>Colloquium</i>
2022	APC Laboratory Paris	<i>Seminar</i>
–	CfA Harvard	<i>Gravitational-wave seminar</i>
–	CCA Flatiron Institute	<i>Gravity seminar</i>
–	MPA, Garching	<i>Seminar</i>
–	Carnegie Observatories	<i>Lunch seminar</i>
–	Nicolaus Copernicus Astron. Center	<i>Colloquium</i>
–	Columbia University	<i>Theoretical High Energy Astrophysics seminar</i>
–	University of Bonn	<i>Stellar seminar</i>
2021	CCA Flatiron Institute	<i>Stars & Compact Objects seminar</i>
–	KU Leuven	<i>Stellar seminar</i>
2020	Kavli IPMU Tokyo	<i>"Astrophysics of LIGO/Virgo sources in O3 era Workshop"</i>
–	Rencontres du Vietnam	<i>"TMEX-2020" Theory meeting experiment conference</i>
–	Geneva Observatory	<i>Seminar</i>
–	Potsdam University	<i>Seminar</i>

AWARDS & GRANTS

2024	MPA Fellowship
2024	Lorentz Center workshop funding grant (~ 30 k€)
2023	”Young Scientist Award” PTA (awarded once every two years for outstanding individual research achievements to a Polish astronomer under the age of 35, received at 30, link)
2023	SSDF grant for a 5-month project of a PhD student under my supervision (~ 10 k€)
2021	GWIC-Braccini Brize (Honorable Mention)
2021	ESO Fellowship
2014, 2015	”Generation of the Future” grant to kick-start a physics students’ lab (~ 200 k€)
2012–2016	Scholarships from the Ministry of Science and Higher Education, Poland (~ 15 k€)
2014	5th place at the International Physicists’ Tournament, EPFL, Lausanne, Switzerland
2012	International Olympiad on Astronomy and Astrophysics, Rio de Janeiro (bronze medal)
2011	International Olympiad on Astronomy and Astrophysics, Katowice (honorable mention)

SUPERVISION & TEACHING

▷ Students:

- Amedeo Romagnolo (NCAC Warsaw/UC San Diego). PhD thesis under my co-supervision: *Gravitational wave sources: Advancements in stellar evolution modeling of binary blackhole progenitors*.
- Debasish Dutta (IIT Palakkad/ISTA). Master project, published in A&A: *Evolutionary nature of puffed-up stripped star binaries and their occurrence in stellar populations*.
- Isabel Kerr (University of St Andrews). Summer student project on MUSE spectroscopy and binary evolution models: *Hide and seek with brawling stellar couples*.
- Michelle Wassink (Radboud/Monash). Master project, published in MNRAS. *Stellar response after stripping as a model for common-envelope outcomes*.
- Simon Weng & Amanda Rubio (ESO). Mentorship & advice for well-being in academia.

▷ Lecturing:

- Massive Binary Evolution and Modeling, PhD level (2023 MESA Summer School – [MESA@Konkoly](#)).
In addition, I led a team of 3 TAs to prepare extensive [Lab Materials](#).
- Stellar Evolution, MSc level (Summer Research Programme 2023 at ESO, a single lecture)
- ESO Summer Research Programme 2023 (ESO). Lecture on basics of stellar evolution, MSc level.

▷ Teaching assistance (Radboud University):

- Advanced Stellar and Binary Evolution (MSc), Gravity and the Cosmis (MSc), Nuclear Evolution of the Universe (BSc), Galaxy Formation and Evolution (MSc)

ACADEMIC SERVICE & CONFERENCE ORGANIZATION

2019–	Referee (<i>ApJL</i> , <i>ApJ</i> , <i>Nature Comm.</i> , <i>A&A</i> , <i>A&A Letters</i> , <i>MNRAS</i> , <i>CQG</i>)	
2025	SOC, PLATO Workshop @ MPA	Garching
2024	SOC chair of a Lorentz Workshop on Stripped Star Discoveries (link)	Leiden
2024–	Reviewer allocation for DPR (data-driven method, development and testing)	ESO Garching
2023	Co-convener of Hot Stars and Binaries splinter session at AG Meeting	Berlin
2023	SOC, Conference @ ESO, <i>Interplay between binaries and star clusters</i>	Garching
2023–	LOC at several conferences and workshops at MPA and ESO	Garching
2023	Scientific Technical Committee (minutes)	ESO Garching
2022	Student Selection Committee	ESO Garching
2022-23	Telescope Time Allocation assistant	ESO Garching
2021-24	Organizer: Journal Club, MESA tutorial sessions	ESO Garching
2019	Co-organizer of the Dutch Astronomy Olympiad	Radboud University
2015	Co-organizer of Conference of Astronomical Student Societies	Izera Park, PL

OUTREACH

2022-now	I co-organize a yearly summer camp stipend for high-school science enthusiasts with a difficult financial situation that we award based on a written article (web link , in Polish)
2022	Outreach talks for visitors at ESO Supernova Center, Germany
2021	Public talk "Astronomy on Tap" in Leiden, Netherlands
2018–2019	Public talks, stargazing events, children festival (Radboud University, Nijmegen)
2012–2019	Outreach talks & entry-level lectures at astronomy youth camps under the "Almukantarat" Astronomical Club in Poland (link , in Eng).
2014–2016	Science Picnics of Polish Radio and Copernicus Science Center, Warsaw, Poland
2015	Outreach talks at local high schools & children summer schools, PL

SCIENTIFIC SOFTWARE & TOOLS

- **Lead developer** of a modern grid-based binary evolution and population synthesis. Application so far: **JK+25, Stegmann & JK 2025**. The first code release in prep.
- **MESA**: 1D stellar evolution code – <http://mesa.sourceforge.net/> – (**JK+19,20,21,22,25**). *Expert user & contributor of open-source tools*: MESA inlist ([link](#)), binaries with MESA ([link](#)).
- **StarTrack**: binary population synthesis code (Belczynski+2002,2008), ideal for gravitational-wave sources. *Code development and extensive usage* (**JK+2017,2018**).
- **StARBURST99**: spectral synthesis code for star-forming galaxies (<https://www.stsci.edu/science/starburst99/docs/default.html>) Limited experience (**JK+2018**).
- **FEWBODY**: toolkit for small-N gravitational dynamics (Fregeau+2004) Good knowledge (**JK+2017**).
- **MOCCA**: stellar cluster Monte Carlo simulator (<https://moccacode.net/>) Limited experience, in collaboration with main developers.

CONSORTIA & INTERNATIONAL COLLABORATIONS

- **BLoEM**: The Binarity at Low Metallicity campaign (currently observing). Large ESO programme monitoring ~ 1000 massive stars in the low-metallicity environment of the SMC.
- **XShootU**: The ULLYSES and X-Shooter massive stars collaboration (data recently taken). Next-generation high-precision stellar astrophysics at low metallicity (1000 Hubble Space Telescope orbits)
- **ET Collaboration**: Einstein Telescope (proposed, ~ 2035) Next-generation ground-based gravitational-wave facility, detecting black hole mergers beyond redshift 10.
- **LISA Consortium**: Laser Interferometer Space Antenna (launch 2035) The first space-based gravitational-wave observatory, opening the mHz frequency window.
- **ENGRAVE**: Electromagnetic counterparts of gravitational-wave sources at the Very Large Telescope.

REFEREED PUBLICATIONS

Publication record:

35 papers (ApJL, A&A Letter, A&A, ApJ, MNRAS, AcA), with **>1520 citations** in total, including **9 first-author publication** with **>400 citations** in total.

First or second authored (>965 citations total)

1. **Stegmann, J. & Klencki, J.**, (2025), ApJL submitted ★ **Co-first authorship** ★ "*Spin-orbit misalignment and residual eccentricity are evidence that neutron star-black hole mergers form from massive triple star evolution*"
2. **Klencki, J.**, Podsiadlowski, P., Langer, N., et al. (2025), A&A submitted "*A fundamental limit to how close binary systems can get via stable mass transfer shapes the properties of binary black hole mergers*"
3. Romagnolo, A., **Klencki, J.**, Vigna-Gómez, A., Belczynski, K. (2025), A&A, 693, A137 "*Development of convective envelopes in massive stars: Implications for gravitational wave sources*"

4. Olejak, A., **Klencki, J.**, Xu, X.-T., Wang, C., Belczynski, K., Lasota, J.-P. (2024), A&A, 689, A305
"Unequal-mass highly spinning binary black hole mergers in the stable mass transfer formation channel"
5. Dutta, D. & **Klencki, J.** (2024), A&A, 687, A215 "On the evolutionary nature of puffed-up stripped star binaries and their occurrence in stellar populations"
6. Costa, G.*, Chruslinska, M.*, **Klencki, J.***, Broekgaarden, F.*, Rodriguez, C.*, Joseph, T.*, Saracino, S.*
(*all authors contributed equally)
"Stellar-Black Holes and Compact Stellar Remnants"
Invited Chapter of the book "Black Holes in the Era of Gravitational Wave Astronomy", edited by Arca Sedda, Bortolas, Spera, to be published by Elsevier (2024)
7. Ramachandran, V., **Klencki, J.**, Sander, A. A. C., Pauli D., Shenar T., et. al. (2023), A&A, 674, L12
"A partially stripped massive star in a Be binary at low metallicity. A missing link towards Be X-ray binaries and double neutron star mergers" [Press Release](#)
8. Kozyreva, A., **Klencki, J.**, Filippenko, A., Baklanov P., Mironov A., et. al. (2022), ApJL, 934, L31
"The Circumstellar Material around the Type IIP SN 2021yja"
9. **Klencki, J.**, Istrate, A. G., Nelemans, G., Pols, O. (2022), A&A, 662, A56
"Partial-envelope stripping and nuclear-timescale mass transfer from evolved supergiants at low metallicity"
10. Blagorodnova, N., **Klencki, J.**, Pejcha, O., Vreeswijk, P. M., Bond, H. E., et al. (2021), A&A, 653, A134
"The luminous red nova AT 2018bwo in NGC 45 and its binary yellow supergiant progenitor"
11. **Klencki, J.**, Nelemans, G., Istrate, A. G., Chruslinska, M. (2021), A&A, 645, A54
"It has to be cool: on supergiant progenitors of binary black hole mergers from common-envelope evolution"
★ top 25 of all A&A articles in 2021, top 5 of those with <10 authors
12. **Klencki, J.**, Nelemans, G., Istrate, A. G., Pols, O. (2020), A&A, 638, A55
"Massive donors in interacting binaries: effect of metallicity"
13. Belczynski, K., **Klencki, J.**, Fields, C., Olejak, A., Berti, E., Meynet, G., et al. (2020), A&A, 636, A104
"Evolutionary roads leading to low effective spins, high masses, and O1/O2 rates of LIGO/Virgo binary black holes"
★ top 10 of all the articles published in 2020 in A&A, MNRAS, and ApJ
14. **Klencki, J.**, Moe, M., Gladysz, W., Chruslinska, M., Holz, D. E., Belczynski, K. (2018), A&A, 619, A77
"Impact of inter-correlated initial binary parameters on double black hole and neutron star mergers"
15. **Klencki, J.**, Wiktorowicz, G., Gladysz, W., Belczynski, K. (2017), MNRAS, 469, 3088
"Dynamical formation of black-hole low-mass X-ray binaries in the field: an alternative to the common envelope"
16. **Klencki, J.**, Wyrzykowski, L., Kostrzewa-Rutkowska, Z., Udalski, A. (2016), Acta Astronomica, 66, 15
"Robust Filtering of Artifacts in Difference Imaging for Rapid Transients Detection"

Co-authored

1. Britavskiy, N., Mahy, L., Lennon, D., Patrick, L., et al. **incl. Klencki, J.** (2025), A&A, 698, A40
"Binarity at LOw Metallicity (BLOeM): Multiplicity of early B-type supergiants in the Small Magellanic Cloud"
2. Hovis-Afflerbach, B., Gotberg, Y., Schootemeijer, A., **Klencki, J.**, Strom, A., Ludwig, B., Drout, M. (2025), A&A, 697, A239 "The mass distribution of stars stripped in binaries: The effect of metallicity"
3. Wang, C., Patrick, L., Schootemeijer, A., de Mink, S. E., et al. **incl. Klencki, J.**, (2025), ApJL, 981, L16 "Using Detailed Single-star and Binary-evolution Models to Probe the Large Observed Luminosity (...)"
4. Preece, H., Vigna-Gómez, A., Rajamuthukumar, A., et al. **incl. Klencki, J.**, (2024), A&A submitted
"The Evolution of Massive Stellar Multiplicity in the Field I. Numerical simulations, long-term evolution (...)"
5. Ramachandran, V., Sander, A., Pauli, D., **Klencki, J.**, et al. (2024), A&A, 692, A90 "X-Shooting ULLYSES: Massive stars at low metallicity: VIII. Stellar and wind parameters of newly revealed stripped stars"
6. Wang, C., Bodensteiner, J., Xu, X.-T., de Mink, S. E., et al. **incl. Klencki, J.** (2024), ApJL, 975, L20
"Stripped Helium Star and Compact Object Binaries in Coeval Populations: Predictions from Binary Evolution"
7. Shenar, T., Bodensteiner, J., Sana, H., Crowther, P., et al. **incl. Klencki, J.** (2024), A&A, 690, A289
"Binarity at LOw Metallicity (BLOeM): A spectroscopic VLT monitoring survey of massive stars in the SMC"
8. Romagnolo, A., Belczynski, K., **Klencki, J.**, Agrawal, P., Shenar T., Szesci, D. (2023), MNRAS, 525, 706
"The role of stellar expansion on the formation of gravitational wave sources"

9. Zak, J., Jones, D., Boffin, H. M. J., Beck, P. G., **Klencki J.**, et al. (2023), MNRAS, 524, 5749
"Everything that glitters is not gold: V1315 Cas is not a dormant black hole"
10. Vink, J., Mehner, A., Crowther, P.A., Fullerton, A., et al. **incl. Klencki, J.** (2023), A&A, 675, A154
"X-Shooting ULLYSES: Massive stars at low metallicity. I. Project description"
11. Pauli, D., Oskinova, L., Hamann, W.-R., Bowman, D., et al. **incl. Klencki, J.** (2023), A&A, 673, A40
"Spectroscopic and evolutionary analyses of the binary system AzV 14 outline paths toward the WR stage (...)"
12. Vigna-Gómez, A., Wassink, M., **Klencki, J.**, Istrate, A., Nelemans, G., et al. (2022), MNRAS, 511, 2326
"Stellar response after stripping as a model for common-envelope outcomes"
13. Belczynski, K., Romagnolo, A., Olejak, A., **Klencki, J.**, Chattopadhyay, D., et al. (2022), ApJ, 925, 69
"The Uncertain Future of Massive Binaries Obscures the Origin of LIGO/Virgo Sources "
14. Wyrzykowski, L., Mroz, P., Rybicki, K., Gromadzki, M. et al. **incl. Klencki, J.** (2020), A&A, 633, A98
"Full orbital solution for the binary system in the northern Galactic disc microlensing event Gaia16aye"
15. Wiktorowicz, G., Wyrzykowski, L., Chruslinska, M., **Klencki, J.**, Rybicki, K., et al. (2019), ApJ, 885, 1
"Populations of Stellar-mass Black Holes from Binary Systems"
16. Rybicki, K., Wyrzykowski, L., **Klencki, J.**, de Bruijne, J., et al. (2018), MNRAS, 476, 2013
"On the accuracy of mass measurement for microlensing black holes as seen by Gaia and OGLE"
17. Chruslinska, M., Belczynski, K., **Klencki, J.**, Benacquista, M. (2018), MNRAS, 474, 2937
"Double neutron stars: merger rates revisited"

Conference proceedings

1. **Klencki, J.**, Nelemans, G. (2019), IAUS Proceedings, 346, 417
"High mass X-ray binaries as progenitors of gravitational wave sources"
2. **Klencki, J.**, Wyrzykowski, L. (2016), Proceedings of the Polish Astronomical Society, 3, 56
"Real-time detection of transients in OGLE-IV with application of machine learning"

REFERENCES

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